



# KTU NOTES

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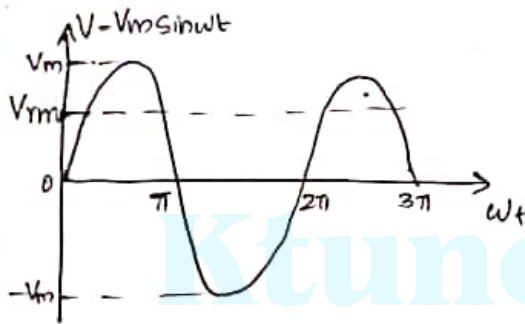
The path for the flow of alternating current is called an AC circuit. This alternating current is used for domestic and industrial purposes. In an AC circuit, the value of the magnitude of current and voltage is not constant. It changes at regular intervals of time.

\* Phasor representation of sinusoidal quantities

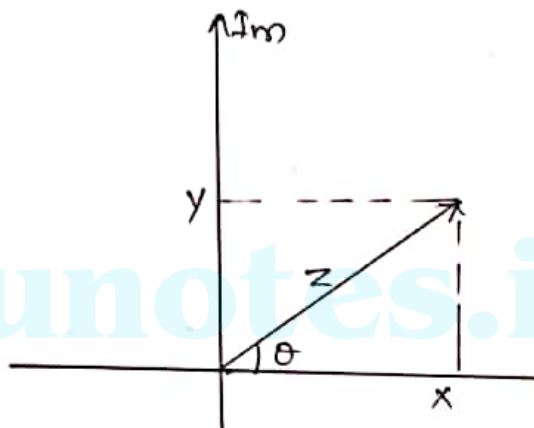
↳ A simple and more direct method of representing sinusoidal quantities is to use the phasor representation. It is easier compared to plotting or analytical method using trigonometric functions.

" A phasor looks like a vector and it rotates at system angular frequency  $\omega$  in positive anticlockwise direction "

Eg: Voltage Waveform and Its Vector.



↳ One complete rotation of a phasor ( $360^\circ$ ) is equivalent to one full cycle of the waveform.



### 1. Rectangular Form

In rectangular form a phasor is represented as

$$Z = x + jy$$

the magnitude of the phasor is,  $Z =$

$$\text{and angle } \theta = \tan^{-1} \frac{y}{x}$$

### 2. Trigonometric Form

In trigonometric form a phasor is represented as

$$Z = r(\cos \theta + j \sin \theta)$$

$$Z = r \angle \theta$$

$$\text{where } r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

#### 4. Exponential form

Here a phasor is represented by,

$$Z = r e^{j\theta}$$

$$\text{where } r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

\*

#### Phasor Addition and Subtraction

Addition and Subtraction of complex numbers can be performed conveniently only if the numbers are in the rectangular form.

The magnitude of  $z = |z| = \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2}$

$$\theta = \tan^{-1} \left( \frac{y_1 + y_2}{x_1 + x_2} \right)$$

Eg:  $V_1 = 2 + 3j$  and  $V_2 = 1 + 2j$

$$V_3 = V_1 + V_2 = (2+1) + (3+2)j = 3 + 5j$$

(ii) Subtraction:  $Z = z_1 - z_2 = (x_1 + jy_1) - (x_2 + jy_2)$   
 $= (x_1 - x_2) + j(y_1 - y_2)$

$$|z| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\theta = \tan^{-1} \left( \frac{y_1 - y_2}{x_1 - x_2} \right)$$

Eg:  $V_1 = 4 + 3j$  and  $V_2 = 2 + 2j$

$$V_3 = V_1 - V_2 = (4-2) + j(3-2) = 2 + j$$

$$\text{then } Z_1 Z_2 = r_1 r_2 \angle \theta_1 + \theta_2$$

$$\text{Eg: } V_1 = 2 \angle 30^\circ \text{ and } V_2 = 6 \angle 20^\circ$$

$$V_3 = V_1 \cdot V_2 = 2 \times 6 \angle (30 + 20) = \underline{\underline{12 \angle 50^\circ}}$$

$$\frac{Z_1}{Z_2} =$$

Division

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2} \angle \theta_1 - \theta_2$$

$$\text{Eg: } V_1 = 6 \angle 30^\circ \text{ and } V_2 = 2 \angle 10^\circ$$

$$V_3 = \frac{V_1}{V_2} = \frac{6}{2} \angle (30 - 10) = \underline{\underline{3 \angle 20^\circ}}$$





$$= \underline{\underline{31.4 + j449.1}}$$

3. Consider voltage phasors  $\vec{V}_1 = 2 + j3$  and represent both voltages in complex plane

- (i) Magnitude of both voltages
- (ii) Sum of  $\vec{V}_1$  and  $\vec{V}_2$
- (iii) Product of  $\vec{V}_1$  and  $\vec{V}_2$

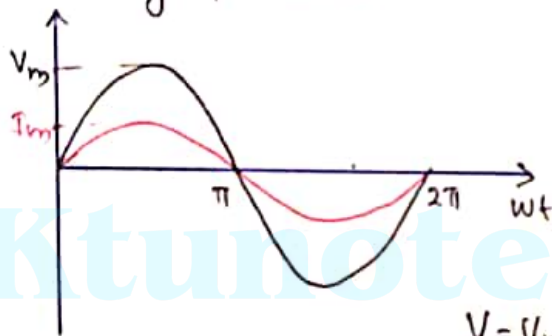
Ans: (i)  $\vec{V}_1 = 2 + j3 \text{ V}$

$$\vec{V}_2 = 1 - j4 \text{ V}$$

(ii) Sum  $\vec{V}_1 + \vec{V}_2 = (2 + j3) + (1 - j4) = \underline{\underline{3 - j}}$

(iii) Product  $\vec{V}_1 \cdot \vec{V}_2 = (\sqrt{13} \angle 56.3^\circ)(\sqrt{17} \angle -76.1^\circ)$

two alternating quantities



$$V = V_m \sin \omega t$$

$$I = I_m \sin \omega t$$

↳ If two quantities are said to be in phase with each other if they reach maximum value or zero cross over point at the same time.

∴ Phase angle difference between  $\vec{V}$  and  $\vec{I}$

↳ One alternating quantity leads another if it reaches maximum value or zero cross over point earlier than the other quantity.

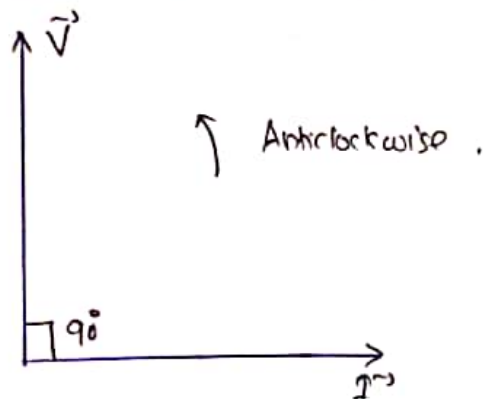
→ Voltage touches zero cross over point at

→ Current reaches zero cross over point

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→ Current lags the voltage by an angle

→ Voltage leads the current by an angle



$$V_p = V_m \sin(\omega t + 90^\circ)$$

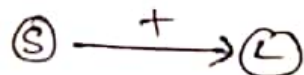
$$I = I_m \sin \omega t$$

$$\vec{V} = V_m \angle 90^\circ$$

$$\vec{I} = I_m \angle 0^\circ$$

Given instant of time.

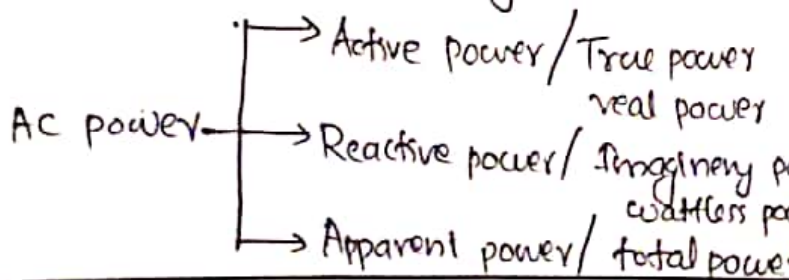
Positive Power :  $\rightarrow$  Power flows from S



Negative Power :  $\rightarrow$  Power flows from L



$\hookrightarrow$  An AC circuit contains resistors inductors and capacitors, as a result the current will lead the voltage by a certain angle or phase angle  $\phi$ . This phase difference should be considered while calculating power



↳ It is the actual or real power consumed at circuit

↳ Unit is watts

↳ Real power is consumed by resistor

$$\therefore \underline{P = I^2 R \text{ watts}}$$

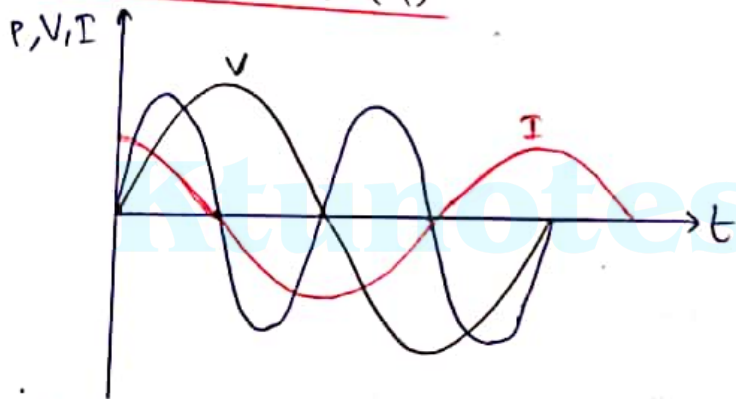
↳ It is the product of voltage and in phase component of current

$$P = VI \cos \phi \text{ watts}$$

↳ V and I are RMS values

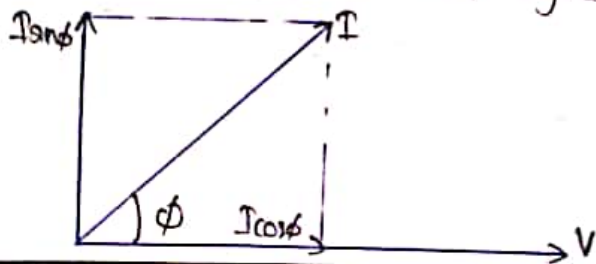
↳  $\phi$  phase angle difference V and I

(ii) Reactive Power (Q)



↳ Reactive power is associated with reactants, capacitors and inductors

↳ Reactive power is stored and released by components in every ac half cycle.



## \* Properties of Reactive Power (Q)

- ↳ It is positive as well as negative
- ↳ It is not used for useful work. It is the power which goes back and forth doing any useful work.

## (ii) Apparent power

- ↳ Apparent power is the combination of active power and reactive power

- ↳ It is the product of RMS values of voltage

$$S = VI \quad \text{VA}$$

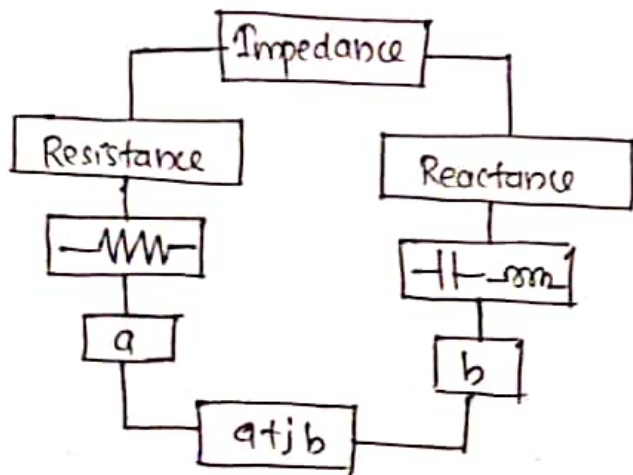
- ↳ Unit is Volt Ampere (VA)

$$S^2 = P^2 + Q^2$$

$$\text{Impedance } Z = \frac{\text{Voltage}}{\text{Current}}$$

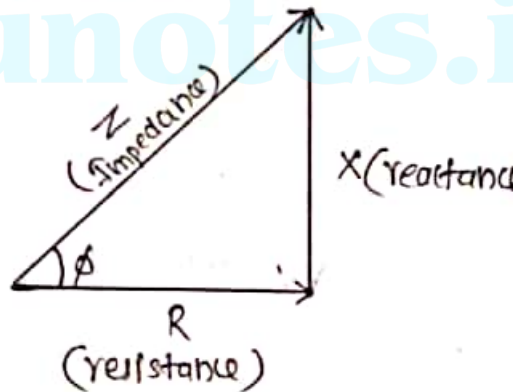
$$\text{Impedance } Z = \text{Resistance} + \text{Reactance}$$

↳ Impedance is the effective resistance of a circuit or component to alternating current from the combined effect of ohmic resistances (offered by C and L)





↳ Resistance ( $R$ ), reactance ( $X$ ), and Impedance ( $Z$ ) can be represented as a right angled triangle called Impedance Triangle.



a, If current lags voltage then power factor  
"lagging P.f"

b, If current leads voltage then power factor  
"leading P.f"

c, If current and voltage are in phase then

(ii) Power factor is the ratio of active power  
apparent power in an ac circuit

$$P.f = \frac{P}{S} = \frac{VI \cos \phi}{VI} = \cos \phi$$

$$P = VI \cos \phi$$

$$Q = VI \sin \phi$$

$$S = VI$$



↳ Value of P.F ranges from 0 to 1

$\cos\phi = 0$  means zero p.f

$\cos\phi = 1$  means unity p.f.

### concept

↳ The total power consumed by an ac circuit has two components

(i) Active power → responsible for doing

(Eg: heating an iron  
Rotating a motor)

(ii) Reactive power → stored and released  
by inductors and capacitors

↳ The value of power factor represents the ratio of total power available for useful work.

is less than 1

Eg: P.f. of motor  $\cos\phi = 0.6$

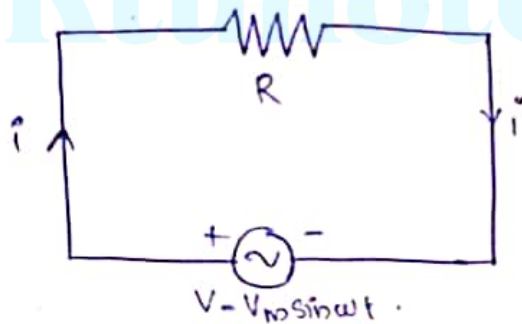
↳ Power factor is less than 1 since motor is  
load

↳ Power factor of an incandescent lamp is  
1 (AC) ( $\approx$  pure resistance)

Resistance of the resistor will be same irrespective of supply frequency.

Let the alternating voltage applied across

$$V = V_m \sin \omega t \quad \text{--- (1)}$$



Instantaneous current ,

$$i = \frac{V}{R} = \frac{V_m \sin \omega t}{R}$$

The value of current will be maximum when

or  $\therefore i = I_m \sin \omega t \quad \text{--- (2)}$

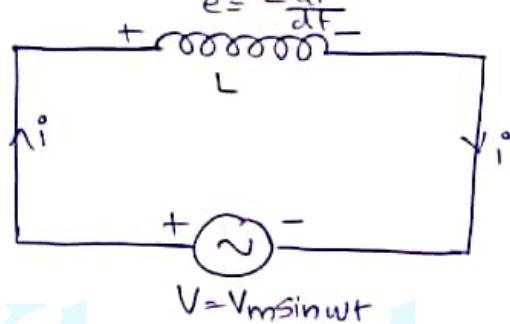
From equation (1) and (3), it is clear that there is no phase difference between applied voltage and current flowing through a purely resistive circuit, i.e. the phase difference between voltage and current is zero. Hence, in a circuit containing pure resistance, current is in phase with the voltage as shown in the above figure.

Instantaneous power  $P = (V_m \sin \omega t)(I_m \sin \omega t)$

$$P = \frac{V_m I_m}{2} 2 \sin^2 \omega t = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} (1 - \cos 2\omega t)$$

$$P = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} - \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos 2\omega t$$

$$P = V_{rms} \cdot I_{rms}$$



↳ When an ac current flows through an coil, an alternating magnetic field is produced according to Faraday's Laws of Electromagnetism. A self induced emf 'e' is developed across the coil.

$$e = -L \frac{di}{dt} \text{ This opposes the applied voltage}$$

The circuit will take some time to come to steady state condition after settling up transient response. Instant the circuit is first switched on.

$$\therefore \text{ we have } u = -e = L \frac{di}{dt} \text{ where } \boxed{U =}$$

$$\text{where } I_m = \frac{V_m}{\omega L} = \frac{V_m}{2\pi f L} = \frac{V_m}{X_L}$$

The term  $\omega L = 2\pi f L$  plays the same resistance in pure resistive ckt and is known as reactance,  $X_L$

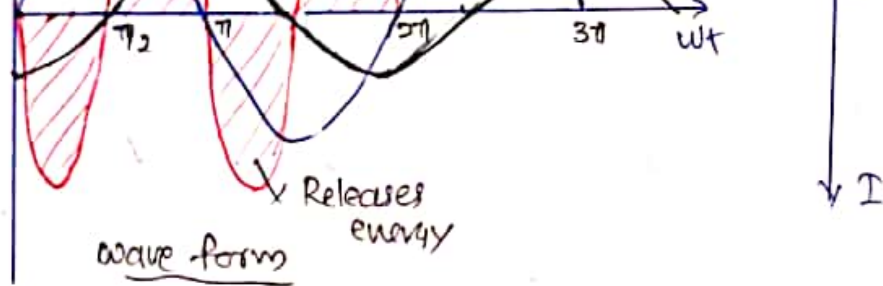
$$\text{i.e. } X_L = \omega L = 2\pi f L \text{ ohms.}$$

where  $L$  = Inductance of coil in Henries

From the equations of  $V$  and  $I$  we can plot

- " In one full cycle an inductor absorbs power. Inductor stores energy for one half cycle and then releases energy for another half cycle
- " Energy is temporarily stored in inductor as

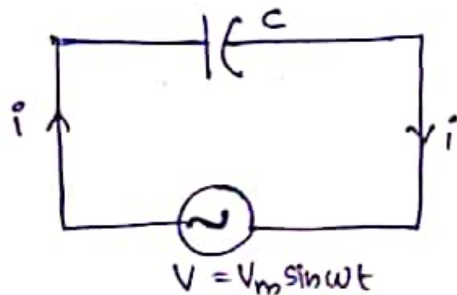




Power factor of a pure inductor  $\cos\phi = 0$   
 It means an inductor deals with only reactive power  $Q$ .

### (iii) AC through a pure capacitor

Consider a pure capacitive ckt as shown below. Let 'c' be capacitance in Henry. An AC source has zero resistance + zero inductance



$$i = \frac{V_m}{(1/\omega C)} \sin(\omega t + \pi/2) = I_m (\sin \omega t + \pi/2)$$

$$i = I_m \sin(\omega t + \pi/2)$$

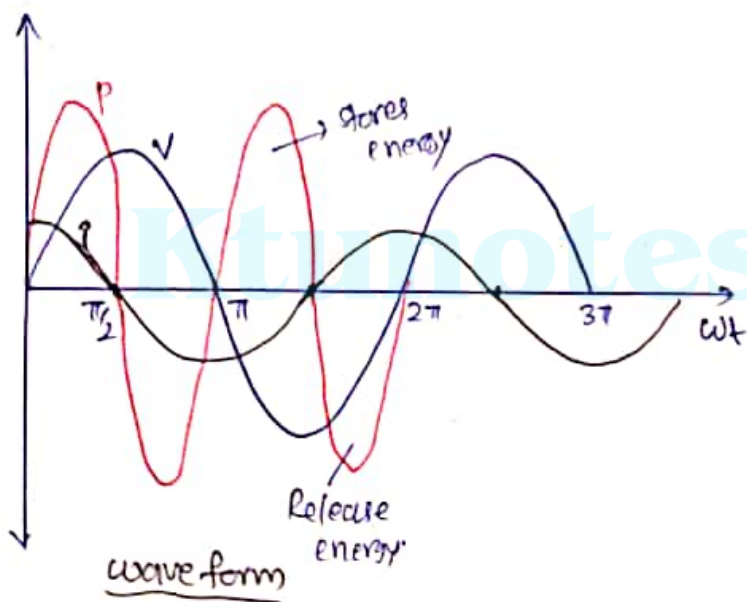
$$\text{where } I_m = \frac{V_m}{(1/\omega C)} = \frac{V_m}{\frac{1}{2\pi f C}} = \frac{V_m}{X_C}$$

the term " $\frac{1}{\omega C} = \frac{1}{2\pi f C}$ " plays the same role as a resistance in pure resistive circuit and is called capacitive reactance ' $X_C$ '

$$\text{i.e., } X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \text{ ohm}$$

where  $C$  = capacitance in Henry

↳ From the equations of  $V$  and  $i$  we can see the waveforms. It is clear that ' $i$ ' leads ' $V$ ' by  $\pi/2$  radian.



### \* Numerical Problems

- 1, A 220V, 50 Hz single phase sinusoidal produces a current of 2.2A in a purely inductive circuit. Determine (i) Inductive reactance of the coil

we have

$$(i) \quad Z = X_L = \frac{V}{i} = \frac{220}{2.2} = \underline{\underline{100 \Omega}}$$

$$X_L = 100 = 2\pi fL = 2 \times \pi \times 50 \times L$$

$$(ii) \Rightarrow L = \frac{100}{2\pi \times 50} = \underline{\underline{0.318 \text{ H}}}$$

(iii) Power absorbed (average power),  $P = 0$  in p

$$(iv) \quad V = V_m \sin \omega t = 220\sqrt{2} \times \sin(2\pi \times 50 t)$$

$$V = \underline{\underline{311.13 \sin(314t) \text{ Volt}}}$$

$$i = I_m \sin(\omega t - \pi/2) = 2.2 \times \sqrt{2} \times \sin(314t - \pi/2)$$

$$i = \underline{\underline{3.11 \sin(314t - \pi/2)}}$$

(in phase with  $I$ )

$$V_L = I X_L = \text{Voltage drop across } L$$

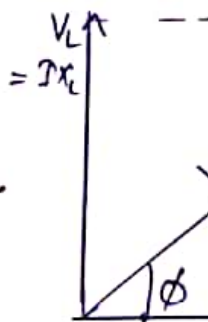
(leads by  $90^\circ$  to  $I$ )

where  $X_L = 2\pi fL = \omega L = \text{Inductive reactance}$

$\rightarrow$  Figure shows the representation of two phasor diagrams.

$\therefore$  The applied voltage ' $V$ ' is the phasor sum of the two voltages.

$$\begin{aligned} V &= \sqrt{V_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2} \\ &= I \sqrt{R^2 + X_L^2} \\ \Rightarrow I &= \frac{V}{\sqrt{R^2 + X_L^2}} = \frac{V}{Z} \end{aligned}$$



where,  $Z = \text{Impedance} = \sqrt{R^2 + X_L^2} = \text{Opposition}$   
series RL circuit to the flow of alternating

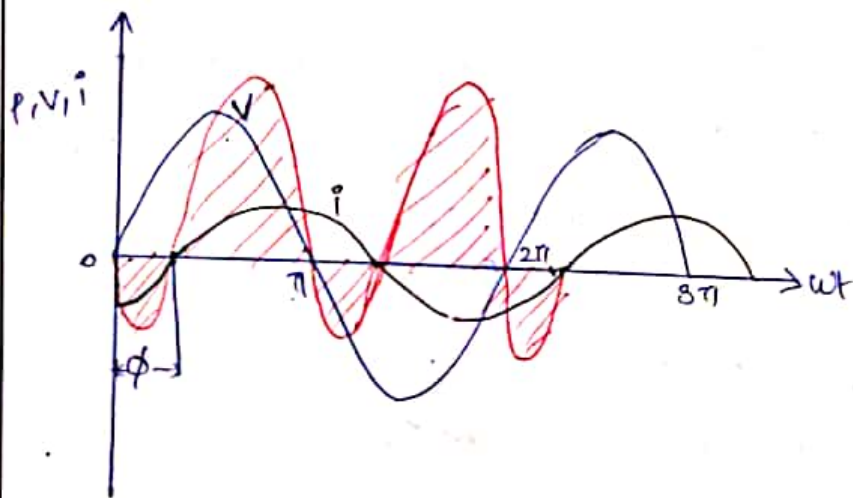
→ If applied voltage is  $V = V_m \sin \omega t$ , the current expression will be,  $i = I_m \sin(\omega t - \phi)$

→ Instantaneous power,  $P = Vi = (V_m \sin \omega t)$

$$\Rightarrow P = \frac{V_m I_m}{2} \sin \omega t \cdot \sin(\omega t - \phi) \quad [\because 2 \sin A \sin B = \cos(A-B) - \cos(A+B)]$$

$$= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} [\cos \phi - \cos(2\omega t - \phi)]$$

$$= V_{rms} \cdot I_{rms} \cdot \cos \phi - \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cdot \cos(2\omega t - \phi)$$



- ↳ From phasor diagram,  $\cos \phi = \frac{I_R}{I} = \frac{V}{V}$
- ↳ Power factor is defined as, the cosine b/w voltage and current in an AC circuit.
- ↳ In this circuit, only power is actually resistance only, Inductance does not consume power
- $$P = VI \cos \phi = (I^2 Z) + \left(\frac{R}{Z}\right) = I^2 R$$

### Numerical

A coil having a resistance of  $10\Omega$  and inductance of  $0.2H$  is connected across a  $200V$ ,  $50Hz$  AC supply. Calculate (i) reactance and impedance of the coil, (ii) current, (iii) phase difference b/w the applied voltage (iv) power factor. Also draw a phasor diagram showing  $V$  and  $I$ .

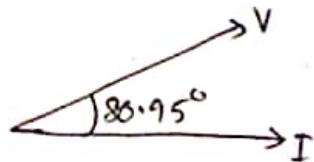
(iii) Phase difference

$$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = 80.95^\circ$$

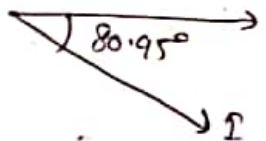
(iv) Power factor,  $\cos \phi = 0.157$  lagging

v) Phasor diagram:-

Current lags behind voltage by an angle



or



2.

The current in series circuit  $R=8\Omega$  and lags the applied voltage by  $70^\circ$ . Determine frequency and the impedance  $Z$ .

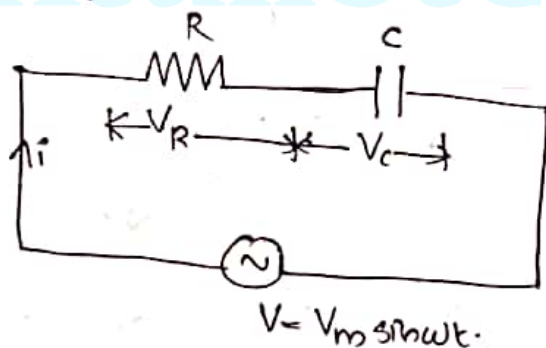
Ans:-  $R=8\Omega$  and  $L=40\text{mH}$ ,  $\phi=70^\circ$

$$\cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + X_L^2}} = \cos 70^\circ = \frac{8}{\sqrt{8^2 + 2\pi f L^2}}$$



## \* AC through Series R-C circuit:

Let an ac supply is given to a series of the R and C, for which a current ' $i$ ' flows in the circuit:



$$V = V_m \sin \omega t$$

$\therefore V_R =$  Voltage drop across resistor  $= iR$

$V_C =$  Drop across capacitor  $= iX_C$

where,  $X_C =$  capacitive reactance  $= \frac{1}{\omega C}$

↳ From the phasor diagram it is clear voltage lags current by an angle  $\phi$ .

↳ From the voltage triangle,

$$V = \sqrt{V_R^2 + V_C^2} = \sqrt{(IR)^2 + (IX_C)^2} \\ = I \sqrt{R^2 + X_C^2} = IZ$$

where  $Z = \sqrt{R^2 + X_C^2}$  → Impedance of

where.  $Z = R + jX_C$

$$\begin{aligned} \text{↳ Again from triangle, } \tan \phi &= \frac{V_C}{V_R} = \frac{X_C}{R} \\ &= \phi = \tan^{-1} \left( \frac{X_C}{R} \right) \end{aligned}$$

power factor of R-C series circuit.

$$\cos \phi = \frac{R}{Z} = \frac{V_R}{V} \text{ lead.}$$

Reactive power  $\phi = VI \sin \phi$  VAR

Apparent power  $S = VI$  VA

↳ In an RC circuit current 'I' leads 'V' by an angle  $\phi$  less than  $90^\circ$ .

\* Numerical

- 1, The waveforms of the voltage and current are given by  $e = 120 \sin(314t)$  and  $i = 10 \sin(314t - 45^\circ)$ . Calculate the values of the resistance, capacitance which are connected in series to form the circuit. Also draw waveforms of current, voltage diagram. Calculate power consumed by the circuit.

$$\therefore I_m = 10 \text{ A} \quad \phi = \pi/6 = 30^\circ$$

$$\therefore I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{10}{\sqrt{2}} = \underline{\underline{7.07 \text{ A}}}$$

$$\text{Now } |Z| = \frac{V}{I} = \frac{89.55}{7.07} = 12.6 \Omega$$

From the expressions  $V$  and  $i$ , it is clear current leads voltage by  $30^\circ$ , so the R-C is capacitive in nature.

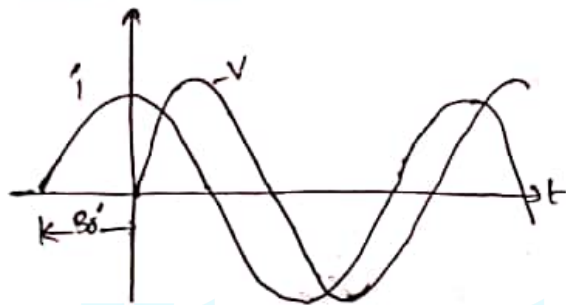
$$\therefore Z = |Z| \angle -\phi = 12.6 \angle -30^\circ = (10.393 - j6) \Omega$$

Comparing with  $Z = R - jX_c$ , we have:-

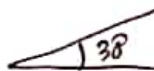
$$\boxed{R = 10.393 \Omega} \quad \text{and} \quad X_c = 6 \Omega =$$

$$\Rightarrow \frac{1}{2\pi f C} = 6$$

$$\therefore C = \frac{6}{2\pi \times 50} = \underline{\underline{530.45 \mu\text{F}}}$$

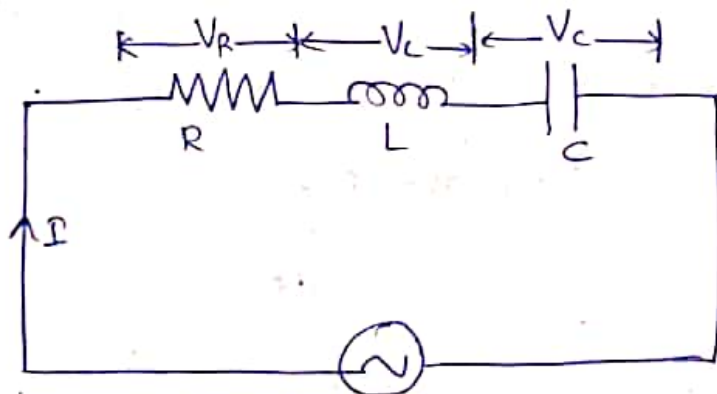


(Waveform)



(Phase)

## \* AC through Series RLC circuit



$$V = V_m \sin \omega t$$

$V_R = IR \rightarrow$  Drop across  $R$  (in phase

$V_L = IX_L \rightarrow$  Drop across 'L' (leading

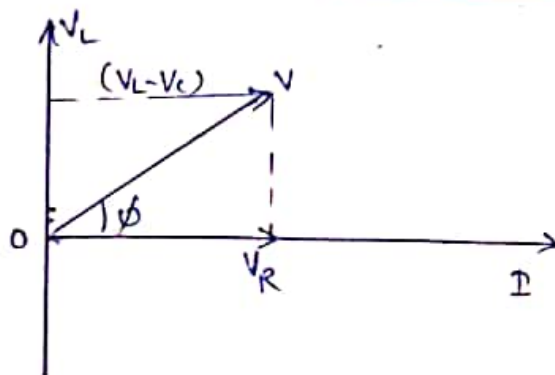
$V_C = IX_C \rightarrow$  Drop across 'C' (lagging

$\rightarrow$  Now the applied voltage can be written

$$\vec{V} = \vec{V}_R + \vec{V}_L + \vec{V}_C$$

$\rightarrow$  The phasor diagram depends on the  
of  $V_L$  and  $V_C$  which ultimately depends on

Case 1: ( $X_L > X_C$ ) (Inductive circuit)



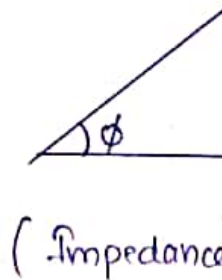
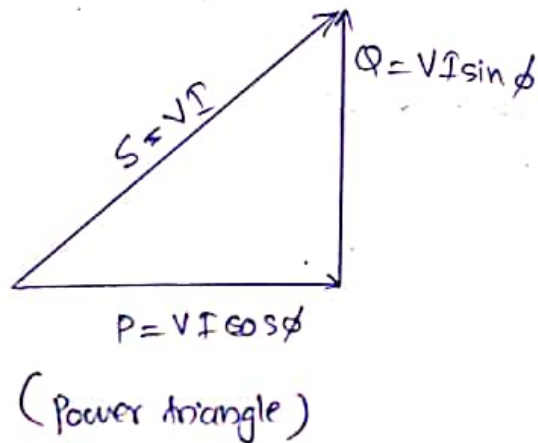
→ In general, the Impedance of RLC is

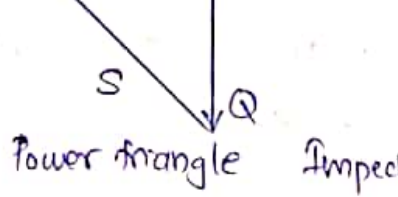
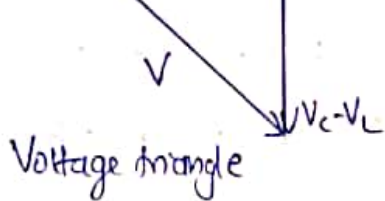
$$Z = R + jX = R + j(X_L - X_C) = |Z| \angle \phi$$

where  $X$  = Net reactance of the circuit

→ Here,  $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$  and  $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$

→ The power factor is given by  $\cos \phi$





case-III ( $X_L = X_C$ ) (Purely resistive circuit):-

↳ Here the net reactance is zero and the circuit behaves as purely resistive circuit. The impedance will be  $\boxed{Z=R}$  and the voltage and current are in phase.

↳ Now, if applied voltage is  $V = V_m \sin(\omega t \pm \phi)$ , the resulting current equation will be,  

$$i = I_m \sin(\omega t \pm \phi).$$

Here '+' sign indicates for capacitive  
 (X<sub>C</sub>)



connected across a 400V, 50 Hz, A.C. Supply combination draws a current of 10A. Find (i) Power factor of the circuit and (ii)

Ans:  $V = 400V$ ,  $f = 50 \text{ Hz}$

$$I = 10 \text{ A}, R = 40 \Omega$$

$$L = 50.07 \text{ mH}$$

We know that

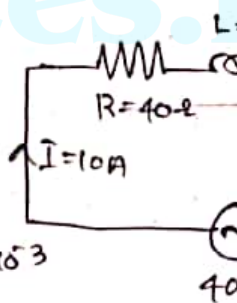
$$X_L = 2\pi fL = 2 \times \pi \times 50 \times 50.07 \times 10^{-3}$$

$$X_L = \underline{\underline{15.73 \Omega}}$$

$$\text{and } X_C = \frac{1}{2\pi fC}$$

$$\text{Now from given data, } |Z| = \frac{|V|}{|I|} = \frac{400}{10} = 40 \Omega$$

$$\therefore Z = R + j(X_L - X_C) \rightarrow |Z| = \sqrt{R^2 + (X_L - X_C)^2}$$



$$\therefore \text{Impedance, } Z = 40 + j(15.73 - 15.73) \\ = 40 + j0 = \underline{\underline{40 \angle 0^\circ}}$$

$$\therefore \text{Power factor, } \cos \phi = \cos 0^\circ = \underline{\underline{1}}$$

\*

### Three phase AC systems

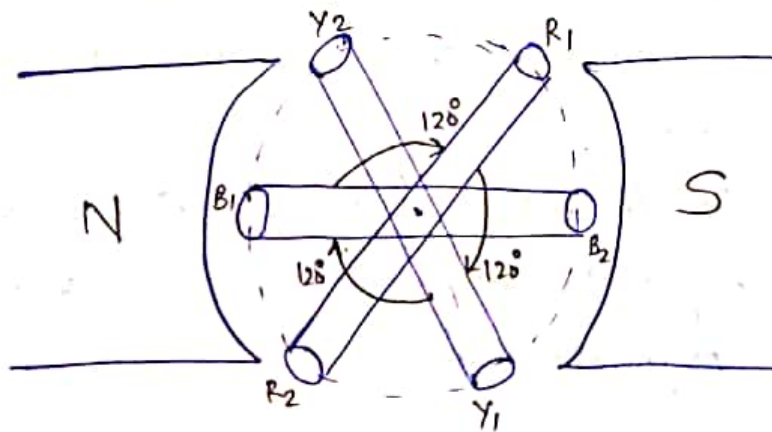
↳ A single phase ac voltage can be rotating a turn made up of two concentric magnetic field. Such a machine is called a single phase alternator.

↳ But practically in many applications, many ac voltages simultaneously such as are called as polyphase system

of  $360^\circ/3 = 120^\circ$  with respect to each other  
three voltages with a phase difference of  
three phase load. Such a supply system  
three phase system.

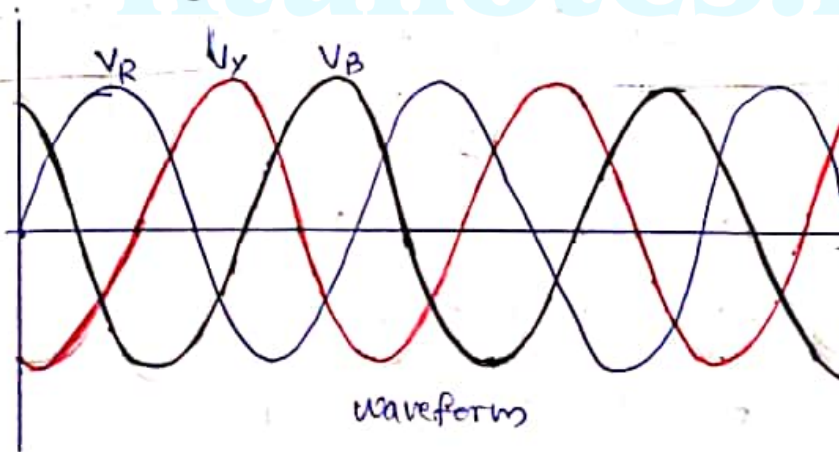
\*

Generation of three phase voltages



from each other.

↳ Let  $V_R$ ,  $V_Y$  and  $V_B$  are the three phase voltages induced in coils  $R_1-R_2$ ,  $Y_1-Y_2$  and  $B_1-B_2$  respectively. All of them are displaced from each other by  $120^\circ$ .



↳ The equations for the three induced

$$V_R = V_m \sin \omega t$$

$$V_Y = V_m \sin(\omega t - 120^\circ)$$

↖  $V_Y$   
→ The sum of all the voltages at any zero.

$$\therefore V_R + V_Y + V_B = V_m \sin \omega t + V_m \sin(\omega t + 120^\circ) + V_m \sin(\omega t + 240^\circ)$$

$$= V_m [\sin \omega t + \sin \omega t \cdot \cos 120^\circ + \cos \omega t \cdot \sin 120^\circ + \sin \omega t \cdot \cos 240^\circ + \cos \omega t \cdot \sin 240^\circ]$$

$$= V_m (\sin \omega t + 2 \sin \omega t \cdot \cos 120^\circ)$$

$$= V_m [\sin \omega t + 2 \sin \omega t \cdot (-\frac{1}{2})]$$

$$= V_m \times 0$$

$$= \underline{\underline{0}}$$

$$\therefore V_R + V_Y + V_B = 0$$

$$\text{ie, } V_R + V_Y = -V_B$$

\* Phase Sequence:

The sequence in which the voltages of the three phases reach their maximum positive value is called phase - sequence. Generally the sequence is R-Y-B.

\* Advantages of 3 phase System:

- (i) Output of three-phase machine is greater than single phase machine. So for same size and voltage, a 3-ph alternator occupies less space and less cost too.
- (ii) Three-phase system needs less copper conducting material than 2 $\phi$  system for same volt-amp.
- (iii) Rotating magnetic field can be produced by 3-phase system.

(vi) Power factor of 1 $\phi$  motors is poor  
three-phase motors.

\*

### Interconnection of three phases

- ↳ In a three phase AC generator, there are three windings. Each winding has two terminals.
- ↳ If a separate load is connected across each winding as shown in the figure below, each phase supplies an independent load through its own pair of wires.
- ↳ Thus six wires will be required to connect a three phase load to a generator.
- ↳ This will make the whole system complicated and costly.

the conductors, the three phase windings of the generator are interconnected.

↳ Interconnection of three phases can be two different ways.

(i) Star connection or wye ( $\Delta$ ) connection

(ii) Delta connection or mesh ( $\Delta$ ) connection

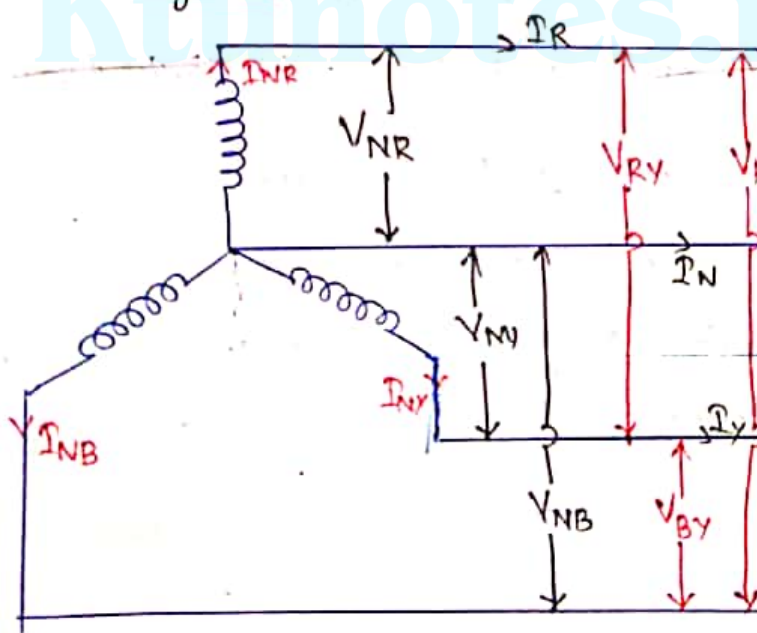
(i) Star connection of three phases

↳ It is a type of connection where each winding is connected to a common point and the remaining three terminals are connected to the circuit.



x

Derivation of relation between line voltages, line and phase currents in connected system.



Phase currents : Current flowing through winding [ $I_{NR}$ ,  $I_{NY}$ ,  $I_{NB}$ ]

Line currents : Current flowing through lines [ $I_R$ ,  $I_Y$ ,  $I_B$ ]

↳ Relation b/w line and phase voltages

↳ If three phase is connected to a balanced load then equal amount of current will flow through each phase

↳ Therefore, phase voltages  $V_{NR}$ ,  $V_{NY}$  are equal in magnitude but displacement angle  $120^\circ$ .

$$|V_{NR}| = |V_{NY}| = |V_{NB}| = V_{ph}$$

$$V_{RY} = V_{NR} + V_{NY}$$

$$V_L = \sqrt{V_{NR}^2 + V_{NY}^2 + 2V_{NR} \cdot V_{NY} \cdot \cos 60^\circ}$$

$$V_L = \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph} \cdot V_{ph} \cdot \cos 60^\circ}$$

$$V_L = \sqrt{2V_{ph}^2 + 2V_{ph}^2 \cdot \cos 60^\circ}$$

$$V_L = \sqrt{3V_{ph}^2}$$

$$V_L = \sqrt{3} V_{ph}$$

$$I_R = I_Y = I_B = I_L$$

$$I_L = I_{ph}$$

$$\text{Power, } P = 3 V_{ph} \cdot I_{ph} \cos \phi$$

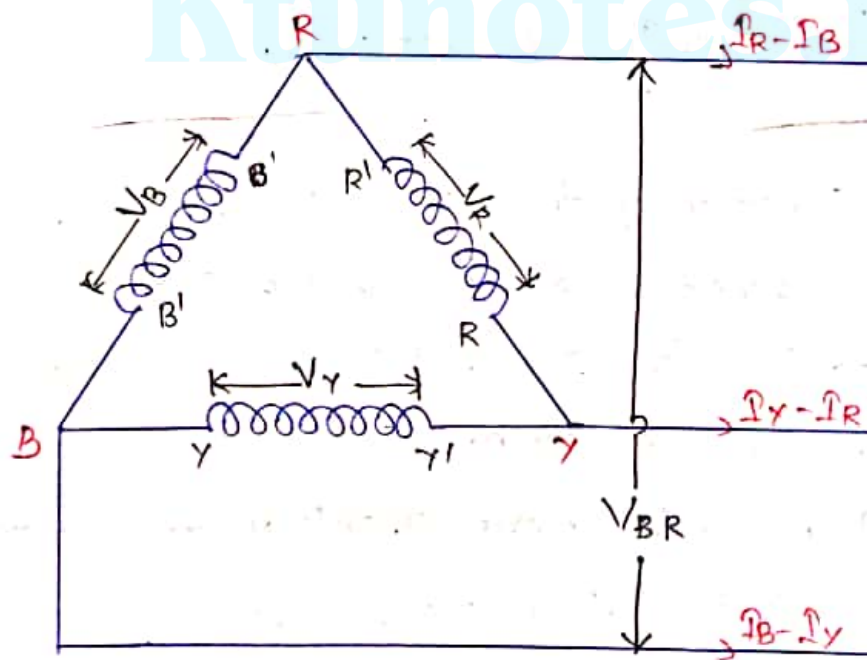
$$P = \sqrt{3} V_L I_L \cos \phi \text{ watts.}$$

(ii) Delta connection of three phases:

↳ In delta connection starting end of one phase is connected to the finishing end of another phase to form a closed mesh.

↳ There is no neutral point in this connection. Therefore it is a three phase three wire system.

- \* Derivation of relation between line voltages, line and phase currents in connected system.



Line quantities :

a) Lin voltages : -  $V_{RY} = V_{YB} = V_{BR}$

$$|I_R| = |I_Y| = |I_B| = I_{ph}$$

$$I_R - I_B = \sqrt{I_R^2 + I_B^2 + 2 I_R I_B \cdot \cos \theta}$$

$$I_L = \sqrt{I_{ph}^2 + I_{ph}^2 + 2 I_{ph} \cdot I_{ph} \cdot \cos 60}$$

$$I_L = \sqrt{3 I_{ph}^2} = \sqrt{3} I_{ph}$$

$$\therefore I_L = \sqrt{3} I_{ph}$$

Power consumed in 3  $\phi$  system,

$$P = 3 V_{ph} I_{ph} \cdot \cos \phi$$

$$= 3 V_L \times \frac{I_L}{\sqrt{3}} \cos \phi$$

$$P = \sqrt{3} V_L I_L \cos \phi \text{ watts.}$$

10 kVA. Find out.

- (i) Line and phase voltages
- (ii) Impedance
- (iii) Reactance / phase, (iv) resistance / phase
- (v) Power factor, (vi) Phase current,
- (vii) Inductance (if frequency is 50 Hz) / phase

Ans: Coils are connected in  $\Delta$  connection

$$I_L = 15 \text{ A}, \quad P_{\text{real}} = 7.5 \text{ kW}, \quad \text{kVA} = 10$$

(i) We know, total power =  $\sqrt{3} V_L I_L$

$$\Rightarrow 10 \times 10^3 = \sqrt{3} V_L \times 15$$

$$V_L = 385 \text{ V} = \underline{\underline{V_{ph}}}$$

$$(i) |Z_{ph}| = \frac{V_{ph}}{I_{ph}} = \frac{385}{(I_L / \sqrt{3})} = \frac{385}{15 / \sqrt{3}} =$$

$\therefore$  We have active power,  $P_{\text{real}} = \sqrt{3} V_L$

(iii)  $X_L \text{ per phase} = X_{Lph} = \underline{\underline{29.41 \Omega}}$

(iv)  $R_{ph} = \underline{\underline{33.33 \Omega}}$

(v) Power factor =  $\underline{\underline{0.75 \text{ lagging}}}$

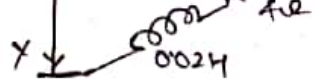
(vi)  $I_{ph} = I_L / \sqrt{3} = 15 / \sqrt{3} = \underline{\underline{8.66 \text{ A}}}$

(vii)  $L_{ph} = \frac{X_{Lph}}{2\pi f} = \frac{29.41}{2 \times \pi \times 50} = \underline{\underline{93.617 \text{ mH}}}$

2. A balanced three phase load consists each having resistance of  $4 \Omega$  and in  $0.02 \text{ H}$ . It is connected to a  $415 \text{ V}$ ,  $50 \text{ Hz}$  AC supply. Determine the phase voltage, current, power factor and active power. loads are connected in
- (i) star      (ii) delta.



$$= \underline{\underline{239.6V}}$$



$$Z_{ph} = \sqrt{R^2 + X_L^2}$$

B. \_\_\_\_\_

$$= \sqrt{R^2 + (2\pi f L)^2} = \sqrt{4^2 + (2\pi \times 50 \times 0.02)^2}$$

$$Z_{ph} = \underline{\underline{7.45 \Omega}}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{239.6}{7.45} = \underline{\underline{32.2 A}}$$

$$\text{Power factor } \cos \phi = \frac{R}{Z} = \frac{4}{7.45} \\ = \underline{\underline{0.538 \text{ lag}}}$$

$$\text{Active power taken } P = 3 V_{ph} \cdot I_{ph} \cdot \cos \phi \\ = 3 \cdot 239.6 \times 32.2 \times 0.538 \\ = \underline{\underline{12.4 \text{ kW}}}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}}$$

B \_\_\_\_\_

$$= \frac{415}{7.45} = \underline{\underline{55.7 A}}$$

$$\text{Power factor} = \frac{R}{Z} = \frac{4}{7.45} = \underline{\underline{0.538 \text{ lag}}}$$

$$\text{Active power } P = 3 V_{ph} I_{ph} \cdot \cos \phi$$

$$P = 3 \times 415 \times 55.7 \times 0.5$$

$$P = \underline{\underline{37.3 \text{ kW}}}$$